

Project Demo Planning

Date	3 July 2026
Team ID	
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	1 Mark

Project Demo Planning:

A well-structured demo plan ensures that the project is presented effectively, covering all key aspects in a clear and organized manner. This document outlines the plan for demonstrating the OptiCrop project, including the flow of the demo and key features to highlight.

S.No	Demo Section	Description	Duration (mins)	Responsible Member
1	Project Introduction	Introduce OptiCrop — what it does, why it matters, and the agricultural problem it solves	2 mins	P Hemanth
2	Dataset & Analysis	Show Crop_recommendation.csv dataset, explain features (N,P,K,temp,humidity,pH,rainfall) and demonstrate univariate & multivariate analysis plots	3 mins	P Hemanth
3	Data Preprocessing	Demonstrate null value check, outlier detection using boxplot, seasonal crop extraction, and train/test split	2 mins	P Hemanth
4	Model Building	Show K-Means Elbow method, Logistic Regression training, classification report with 94% accuracy, and model saving as model.pkl	3 mins	P Hemanth
5	Flask App Demo	Run app.py, open browser, show Home page, About page, then enter test values (N=105,P=35,K=40,T=25,H=64,pH=7,R=160) and demonstrate crop prediction output	3 mins	P Hemanth
6	Conclusion & Summary	Summarize key achievements, model accuracy, features implemented, and future scope of OptiCrop	2 mins	P Hemanth

Demo Flow Summary:

Step	Activity	Notes
1	Introduction & Problem Statement	Explain agricultural crop selection problem and how OptiCrop solves it using ML
2	Solution Overview	Walk through the project architecture: Dataset → Preprocessing → ML Model → Flask App
3	Live Feature Demonstration	Run Jupyter notebook cells + Run app.py + Show prediction result in browser
4	Q&A; Session	Answer any questions about the ML model, accuracy, or Flask application functionality